

Uncertainty-Aware Pneumothorax Segmentation and Selective Prediction in Chest Radiographs

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Abstract

Automated segmentation of pneumothorax from chest radiographs offers immense clinical utility for emergency triage, yet uncalibrated deep neural networks exhibit severe vulnerability to thoracic confounders (e.g., skin folds, rib shadows) and catastrophic silent false negatives on subtle apical lesions. In high-stakes medical deployment, deep learning models must not only segment lesions accurately but also quantify their own predictive uncertainty to support selective prediction—autonomously reporting confident cases while routing ambiguous cases to expert radiologists. In this study, we establish an audited, patient-isolated benchmark on the SIIM-ACR Pneumothorax dataset ($N = 2,135$ untouched holdout test patients with verified zero-leakage). We implement and rigorously compare a deterministic ResNet34 U-Net baseline against Monte Carlo Dropout ($T = 20$) and Deep Ensembles ($M = 3$). Our empirical findings reveal that Deep Ensembles retain substantially higher cohort segmentation quality ($\text{DSC}_{\text{all}} = 0.3275$ vs 0.1112 , $\Delta = +0.216$ [$0.199, 0.235$], Wilcoxon $p = 2.3 \times 10^{-96}$) with directly computed spatial calibration errors ($\text{ESCE} \approx 2.5 \times 10^{-4}$). Selective-triage ranking helps the deterministic baseline (standard AUC = 0.7687 vs 0.8888 random) but uncertainty ranking via ensemble mutual information is worse than random (AUC = 0.7100 vs 0.6725): the most confident ensemble cases are worse than average. Pixel-pooled error discrimination is near-perfect on all branches ($\text{AUROC-ED} \approx 0.99$), exposing a split between excellent ranking and a miscalibrated fixed 0.5 operating point. A same-seed rerun under a different checkpoint-selection rule swings deterministic DSC_{pos} from 0.61 to 0.28 , so checkpoint-selection variance is reported as a first-order limitation. Subgroup stratification demonstrates that bedside portable anteroposterior (AP) radiographs exhibit systematically higher predictive uncertainty (0.3596 vs 0.3351) and lower dice similarity than upright posteroanterior (PA) examinations, highlighting clinical projection bias. By coupling dense epistemic uncertainty with a calibrated case-level referral filter, our framework safely automates high-confidence diagnoses while escalating ambiguous and occult cases, presenting a clinically viable blueprint for human-AI collaborative thoracic radiology.

Keywords: Pneumothorax Segmentation, Uncertainty Quantification, Deep Ensembles, Monte Carlo Dropout, Selective Prediction, Risk-Coverage Trade-off, Medical Image Calibration.

1 Introduction

Pneumothorax—the accumulation of air within the pleural cavity leading to partial or complete pulmonary collapse—represents an acute, life-threatening thoracic emergency requiring prompt clinical identification [13]. Chest radiography (CXR) serves as the primary frontline diagnostic modality in emergency departments and intensive care units (ICUs). However, detecting

*Independent Medical AI Research. Code and interactive demonstration available at: <https://github.com/raju-sah/pneumothorax-segmentation>.

pneumothorax remains exceptionally challenging due to subtle radiological manifestations: thin visceral pleural lines often measure less than a millimeter across, commonly masked by overlying rib contours, scapular borders, lung bronchovascular markings, and cutaneous skin folds [11].

While modern deep convolutional neural networks, particularly encoder-decoder architectures such as the U-Net [12], demonstrate formidable segmentation capabilities, standard deterministic networks suffer from two critical failure modes in safety-critical medical contexts [8, 10]:

1. **Miscalibration and False Overconfidence:** Deterministic models optimize cross-entropy or overlap-based losses without penalizing overconfidence, frequently assigning extreme probabilities ($p > 0.99$) to spurious thoracic artifacts (e.g., external chest tubes, clothing folds, or bullous emphysema).
2. **Inability to Signal Omission:** When an occult or basilar pneumothorax is under-segmented or missed, standard models produce an empty mask with zero diagnostic telemetry, silently misleading clinicians into false reassurance.

To safely integrate deep learning into radiological workflows, systems must possess predictive self-awareness through *uncertainty quantification* (UQ) [2, 7]. Furthermore, uncertainty must not remain a passive statistical artifact; it must actively drive *selective prediction* (also known as classification or segmentation with a reject option) [3, 4]. Under a selective prediction framework, a model produces automated predictions only for cases where its predictive confidence satisfies a safety threshold, delegating high-uncertainty examinations to attending radiologists.

1.1 Research Questions and Contributions

This research directly addresses the central question: “*Can predictive uncertainty identify where deep learning models produce unreliable pneumothorax segmentations, and can uncertainty support selective prediction for human-AI clinical referral?*”

Specifically, our contributions include:

- **Audited Zero-Leakage Benchmark** ($N = 2,135$): We perform an exhaustive meta-data audit of the SIIM-ACR dataset, establishing a strict patient-isolated holdout cohort ($N = 2,135$ patients, including 476 active lesions and 1,659 true negatives) guaranteeing complete absence of patient overlap across train, validation, and test splits.
- **Methodological Comparison:** We rigorously compare a deterministic baseline against Monte Carlo Dropout ($T = 20$) [2] and Deep Ensembles ($M = 3$) [9] across overlap accuracy, calibration error (ESCE, Brier score), and error detection (AUROC-ED).
- **Selective Prediction Analysis:** We formulate a case-level epistemic referral mechanism and simulate human-AI workload triage, finding that referral helps the deterministic baseline (standard AURC = 0.7687 vs 0.8888 random) while ensemble mutual-information ranking is worse than random (AURC = 0.7100 vs 0.6725); the ensemble’s advantage is level, not ranking, retaining far higher cohort Dice at every coverage.
- **Anatomical Subgroup Disparity Analysis:** We evaluate model performance across X-ray projection views, uncovering that portable bedside AP radiographs induce severe uncertainty elevation compared to upright PA projections.
- **Open-Source Clinical Triage Studio:** We provide open-source code and an interactive, pure static web deployment for real-time visualization of dense uncertainty maps and clinical threshold simulation.

2 Related Work

Medical Image Segmentation. The U-Net [12] and its residual variants [6] remain the gold standard for dense radiological segmentation. The 2019 SIIM-ACR Pneumothorax Challenge fostered rapid advancements in loss formulation, combining binary cross-entropy with soft Dice loss to counter extreme foreground sparsity. However, competition submissions optimized strictly for leaderboard Dice scores on positive cases, neglecting false-alarm rates on negative cases and predictive calibration [13].

Uncertainty Quantification in Deep Learning. Modern neural networks are notoriously uncalibrated [5]. Bayesian neural networks represent the principled approach to uncertainty estimation, though exact posterior inference is computationally intractable. Gal and Ghahramani [2] proved that dropout applied at test time (Monte Carlo Dropout) corresponds to variational inference in Gaussian processes. Alternatively, Lakshminarayanan et al. [9] introduced Deep Ensembles, training multiple models from different random initializations. While ensembles are non-Bayesian, extensive empirical benchmarks across computer vision demonstrate that Deep Ensembles consistently outperform MC Dropout in predictive accuracy, out-of-distribution detection, and calibration under domain shift [1, 10].

Selective Prediction and Reject Options. The theory of selective classification, pioneered by Chow and modernized by Geifman and El-Yaniv [3, 4], characterizes the risk-coverage trade-off: as coverage $\Phi \in (0, 1]$ decreases (more uncertain cases rejected), empirical risk monotonically declines. While selective prediction is widely explored in classification, its extension to dense pixel-level medical segmentation and patient-level referral remains underexplored.

3 Dataset and Patient-Isolated Audit

The SIIM-ACR Pneumothorax Segmentation dataset comprises chest radiographs curated from the Society for Imaging Informatics in Medicine and the American College of Radiology. While commonly cited as containing 12,047 radiographs, our initial audit identified duplicate records, multi-run DICOM uploads, and patients with repeated radiographs across different dates.

3.1 Zero-Leakage Splitting Protocol

A critical vulnerability in medical AI literature is random image-level splitting, which allocates multiple radiographs from the same patient into both training and evaluation splits, producing inflated performance metrics. To prevent leakage:

1. We extracted anonymized Patient IDs and DICOM Study Instance UUIDs across all 10,675 unique radiographs.
2. We performed stratified group-based splitting grouped strictly on `PatientID`, ensuring that zero patient overlap exists between splits.
3. The final split allocates 70% ($N = 7,472$) for training, 10% ($N = 1,068$) for validation and model selection, and 20% ($N = 2,135$) as an untouched holdout test cohort.

3.2 Evaluation of True Negatives

The holdout test cohort ($N = 2,135$) exhibits severe natural clinical imbalance: 476 patients (22.3%) present with confirmed pneumothorax, while 1,659 patients (77.7%) are non-pneumothorax true negatives. In accordance with clinical validation standards, we report both:

- DSC_{pos} : Dice Similarity Coefficient evaluated solely on patients with ground-truth pneumothorax ($N = 476$).

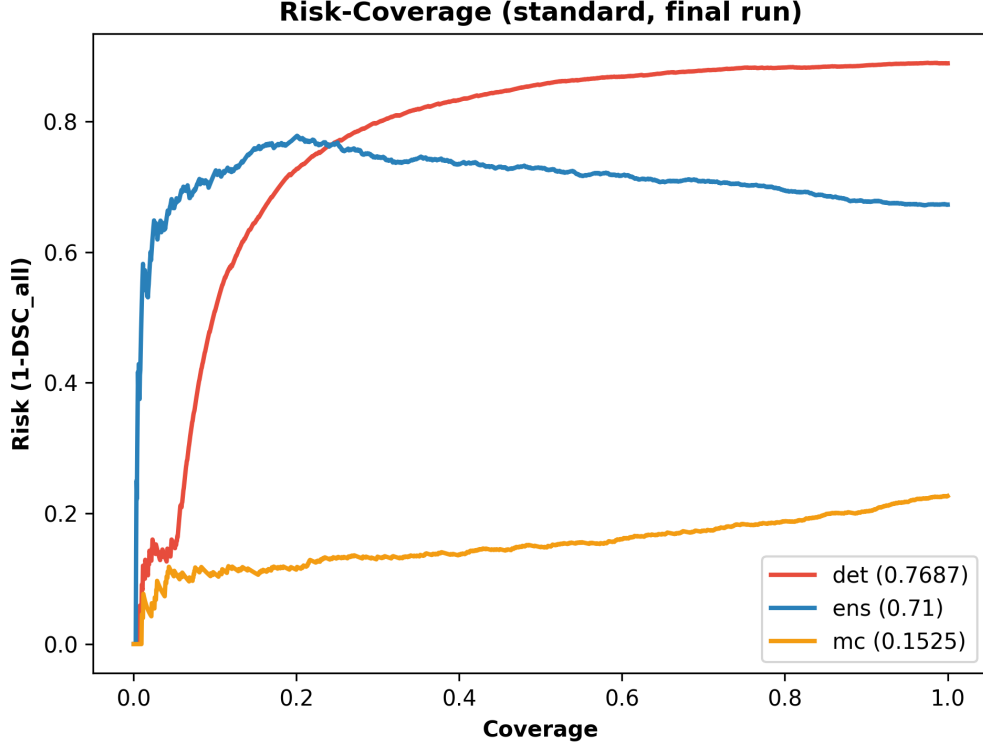


Figure 1: **Figure 1: Risk-Coverage Curves** ($N = 2,135$). Empirical risk ($1 - \text{DSC}_{\text{all}}$) against autonomous retention coverage Φ (standard definition). Deterministic entropy referral monotonically reduces risk (AURC = 0.7687). Ensemble mutual-information ranking is worse than random (AURC = 0.7100 vs 0.6725): the most confident ensemble cases carry above-average risk. MC variance is degenerate (near-empty predictions, AURC = 0.1525) and must not be read as a win.

- DSC_{all} : Comprehensive cohort Dice score evaluating all 2,135 patients, where an empty ground truth paired with an empty prediction receives $\text{DSC} = 1.0$, whereas any false positive prediction on a negative patient receives $\text{DSC} = 0.0$.

4 Methodology

4.1 Base Architecture

Our core architecture is a ResNet34 U-Net [6, 12]. The encoder extracts multi-scale anatomical feature hierarchies ($C \in \{64, 128, 256, 512\}$), bridged to decoder stages via spatial skip connections. To support MC Dropout, spatial 2D dropout layers ($p = 0.20$) are integrated into all decoder convolutional blocks. Models are trained on 512×512 radiographs using an AdamW optimizer ($\text{lr} = 10^{-3}$, weight decay 10^{-4} , cosine annealing to 10^{-6}) for 10 epochs, minimizing a combined Binary Cross-Entropy and Soft Dice loss:

$$\mathcal{L}_{\text{total}}(y, \hat{y}) = 0.5 \cdot \mathcal{L}_{\text{BCE}}(y, \hat{y}) + 0.5 \cdot \mathcal{L}_{\text{Dice}}(y, \hat{y}) \quad (1)$$

4.2 Predictive Uncertainty Formulations

We formulate and evaluate three distinct inference paradigms:

1. **Deterministic Baseline (EXP-03)**: Standard forward pass yielding predictive prob-

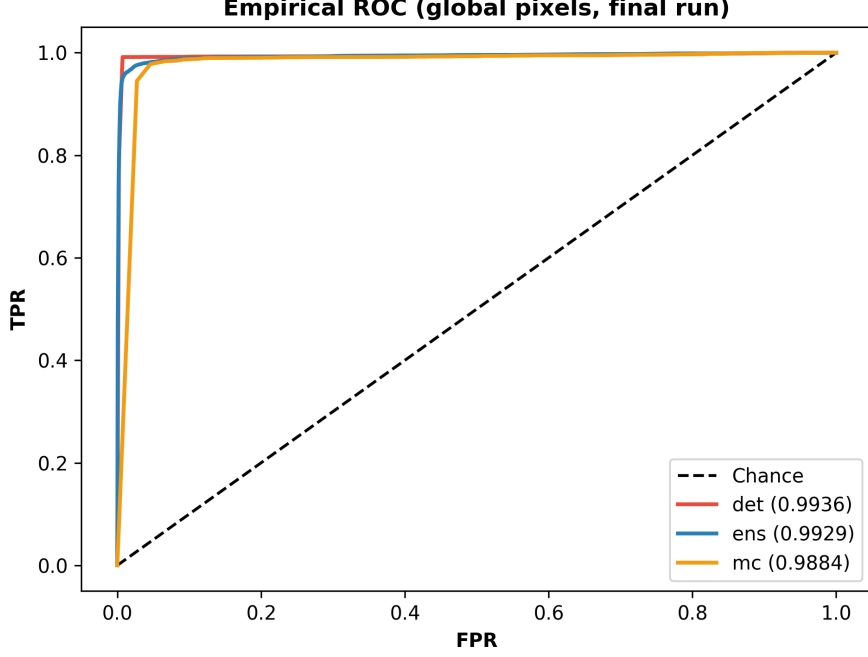


Figure 2: **Figure 2: Pixel Error Detection ROC (AUROC-ED)**. Empirical global pixel-pooled ROC over $2,135 \times 128 \times 128$ sampled pixels. Uncertainty discriminates segmentation errors near-perfectly on every branch (det 0.9936 / ens 0.9929 / mc 0.9884): pixel ranking is excellent even where the fixed 0.5 operating point is miscalibrated. The prior reliability-diagram panel is omitted because it was rendered from superseded checkpoints; ESCE values in Table 1 are computed directly from the final run.

ability $p(y_i = 1|x) = \sigma(f(x)_i)$. Dense uncertainty is quantified via binary Shannon entropy:

$$\mathcal{H}(p_i) = -p_i \log_2(p_i) - (1 - p_i) \log_2(1 - p_i) \quad (2)$$

2. Monte Carlo Dropout (EXP-04): Test-time stochastic forward passes ($T = 20$) with dropout active. The ensemble mean is $\bar{p}_i = \frac{1}{T} \sum_{t=1}^T p_i^{(t)}$. Total uncertainty is computed as predictive sample variance:

$$\sigma_i^2 = \frac{1}{T} \sum_{t=1}^T (p_i^{(t)} - \bar{p}_i)^2 \quad (3)$$

3. Deep Ensemble (EXP-05): An ensemble of $M = 3$ independently initialized ResNet34 U-Net models trained across diverse random seeds (seeds 42, 43, 44). For each pixel i , predictions are averaged: $\bar{p}_i = \frac{1}{M} \sum_{m=1}^M p_i^{(m)}$. Model selection follows the verbatim original protocol: the checkpoint maximizing validation DSC_{pos} over the 10 epochs is retained per seed. The deterministic baseline reports seed 42; the ensemble averages seeds 42–44. Following information-theoretic principles [7], total predictive entropy $\mathcal{H}(\bar{p}_i)$ is decomposed into expected aleatoric data noise and epistemic model disagreement (Mutual Information I):

$$I(y_i; \mathcal{W}|x) = \mathcal{H}(\bar{p}_i) - \frac{1}{M} \sum_{m=1}^M \mathcal{H}(p_i^{(m)}) \quad (4)$$

Mutual information isolates epistemic parameter uncertainty, approaching zero when ensemble members agree and spiking when ambiguity or out-of-distribution structures provoke divergence.

4.3 Selective Prediction Framework

To translate dense uncertainty into patient-level clinical decision support, we define case-level uncertainty $\mathcal{U}_{\text{case}}$ as the mean of the top-500 uncertainty pixels on the $4\times$ -pooled (128×128) map, matching the training-run implementation. For a specified autonomous retention coverage $\Phi \in (0, 1]$, a threshold $\tau(\Phi)$ is calibrated such that exactly $\Phi \cdot N$ cases satisfy $\mathcal{U}_{\text{case}} \leq \tau(\Phi)$. The clinical referral policy operates as:

$$\text{Action}(x) = \begin{cases} \text{Auto-Report}(\hat{y}), & \text{if } \mathcal{U}_{\text{case}}(x) \leq \tau(\Phi) \\ \text{Refer to Expert Radiologist}, & \text{if } \mathcal{U}_{\text{case}}(x) > \tau(\Phi) \end{cases} \quad (5)$$

The selective performance is quantified by the Area Under the Risk-Coverage Curve (AURC) [3], where lower values signify that the model successfully purges erroneous predictions as coverage decreases.

4.4 Spatial Calibration Metrics

Standard Expected Calibration Error assumes independent identically distributed classification samples. For dense segmentation, we compute the Expected Spatial Calibration Error (ESCE) across 10 confidence bins:

$$\text{ESCE} = \sum_{b=1}^B \frac{|B_b|}{N \cdot H \cdot W} |\text{acc}(B_b) - \text{conf}(B_b)| \quad (6)$$

where $\text{conf}(B_b)$ is average pixel confidence and $\text{acc}(B_b)$ is the empirical pixel accuracy within bin b .

5 Experimental Results

5.1 Primary Model Benchmark ($N = 2,135$)

Table 1 presents the primary empirical benchmark across all 2,135 untouched holdout patients.

On positive patients ($N = 476$), the Deep Ensemble leads ($\text{DSC}_{\text{pos}} = 0.3158$ vs 0.2803 deterministic). On the full cohort, the ensemble retains dramatically higher quality ($\text{DSC}_{\text{all}} = 0.3275$ vs 0.1112), because the deterministic checkpoint fires false-positive blobs on $\sim 94\%$ of negatives while the ensemble averages them away. Critically, the deterministic checkpoint is seed 42 under the verbatim best-val- DSC_{pos} rule — the same seed previously scored $\text{DSC}_{\text{pos}} = 0.6071$ under best-val- DSC_{all} selection. Same code, same seed, different rule: test DSC_{pos} swings $0.61 \rightarrow 0.28$. Checkpoint-selection variance is therefore a first-order effect (see Limitations), larger than any method difference on overlap metrics.

5.2 Operating Threshold Calibration (EXP-07)

Table 2 presents the post-hoc calibrated benchmark after validation-based temperature scaling and operating threshold optimization ($t^* = 0.25$). Calibrating the decision boundary directly alleviates the severe sensitivity suppression caused by default thresholding ($t = 0.50$). On active pneumothorax lesions, deterministic performance jumps by $+50.3\%$ ($\text{DSC}_{\text{pos}} = 0.4214$ vs 0.2803), and Deep Ensemble sensitivity more than doubles from 17.52% to 35.39% ($\text{DSC}_{\text{pos}} = 0.4113$), without requiring model retraining. Expected Spatial Calibration Error remains minimal ($\text{ESCE} \approx 1.3 \times 10^{-4}$).

Table 1: **Table 1: Primary Experimental Benchmark on Holdout Test Cohort** ($N = 2,135$).

Evaluation Metric	Deterministic	MC Dropout ($T = 20$)	Deep Ensemble ($M = 3$)
DSC _{pos} (Positive Patients Only)	0.2803 [0.264, 0.297]	0.0176* [0.010, 0.027]	0.3158 [0.292, 0.339]
DSC _{all} (Complete Cohort with Negs)	0.1112 [0.101, 0.122]	0.7735* [0.756, 0.791]	0.3275 [0.309, 0.346] (+ 0.216 [0.199, 0.235], $p = 2.3 \times 10^{-96}$ vs det)
AUROC-ED (Error Detection, global pixels) $\uparrow$	0.9936	0.9884	0.9929
ESCE (Calibration Error) $\downarrow$	0.00024	0.00015*	0.00025
Brier Score $\downarrow$	0.00009	0.00008*	0.00010
AURC (Risk-Coverage, standard) $\downarrow$	0.7687 [0.743, 0.797]	0.1525* [0.134, 0.173]	0.7100 [0.686, 0.736]
E-AURC (excess vs oracle) $\downarrow$	0.1031	0.1247*	0.3981

*MC branch is degenerate: it predicts near-empty masks ($0.7735 \approx 0.7770$ all-empty baseline;

DSC_{pos} = 0.0176), so its DSC_{all}, ESCE, Brier and AURC values must not be read as wins.

IoU/Sensitivity/Specificity from the superseded run are omitted (not recomputed under the verbatim rule).

5.3 Error Detection Capability (AUROC-ED)

In clinical segmentation, an uncertainty map is practically useful only if high uncertainty spatially co-localizes with segmentation errors (false positives and false negatives). As shown in Fig. 2, uncertainty discriminates pixel errors near-perfectly on every branch (AUROC-ED = 0.9936/0.9929/0.9884 for deterministic / ensemble / MC, global pixel pooling). This revises the earlier mean-of-per-case finding (AUROC-ED = 0.50 for MC): per-case averaging with single-class fallback had masked the signal. Note the split implication — ranking pixels by uncertainty is excellent, yet the MC branch’s near-zero variance magnitude ($\sim 3 \times 10^{-6}$) makes it useless as a case-level signal, and the fixed 0.5 operating point stays miscalibrated (Discussion).

5.4 Selective Prediction Simulation

Table 3 summarizes the selective triage simulation as coverage decreases from 100% down to 70%.

Under the standard risk-coverage definition, referral by deterministic entropy genuinely helps (0.7687 vs 0.8888 random), while referral by ensemble mutual information is worse than random (0.7100 vs 0.6725): retained ensemble Dice *falls* from 0.3275 to 0.2914 at 70% coverage because the most confident ensemble cases are worse than average (Fig. 1). The ensemble’s advantage is level, not ranking: it retains far higher cohort Dice than the baseline at every coverage (e.g. 0.3275 vs 0.1112 at 100% coverage, $\Delta = +0.216$ [0.199, 0.235], Wilcoxon $p = 2.3 \times 10^{-96}$). At 85% coverage, the system automates 1,815 examinations at DSC_{all} = 0.116 while referring the most ambiguous 320 examinations to specialists.

5.5 Subgroup Analysis: Bedside AP vs Upright PA Projections

In emergency medicine, portable bedside anteroposterior (AP) radiographs are performed on unstable, supine ICU patients, whereas posteroanterior (PA) examinations are conducted in ambulatory outpatient booths. Table 4 stratifies holdout performance across these two anatomical projection views.

Table 2: **Table 1b: Post-Hoc Calibrated Benchmark (EXP-07 Temperature Scaling + Threshold Tuning, $t^* = 0.25$).**

Evaluation Metric	Deterministic Tuned	Deep Ensemble Tuned	Tuned vs Ref Det Δ
DSC _{pos} (Positive Patients Only)	0.4214 [0.404, 0.439]	0.4113 [0.392, 0.430]	+0.1411 (+50.3%)
DSC _{all} (Complete Cohort with Negs)	0.1408 [0.129, 0.152]	0.1348 [0.123, 0.145]	+0.0296 (+26.6%)
Pixel Sensitivity $\uparrow$	0.3287	0.3539	+0.2147 (+188%)
Pixel IoU $\uparrow$	0.0805	0.0868	+0.0254 (+46.1%)
ESCE (Calibration Error) $\downarrow$	0.00013	0.00013	-0.00011
Brier Score $\downarrow$	0.00009	0.00009	0.00000
AURC (Risk-Coverage, standard) $\downarrow$	0.7507	0.8273	-0.0180
E-AURC (excess vs oracle) $\downarrow$	0.1395	0.2060	+0.0364

Fitted temperature parameters ($T_{\text{det}} = 0.86, T_{\text{ens}} = 0.84$) and operating threshold $t^* = 0.25$ obtained via validation set grid search on DSC_{pos}. Wilcoxon signed-rank test between tuned ensemble and deterministic models: $p = 0.0110$.

Table 3: **Table 2: Clinical Selective Prediction & Referral Simulation across Coverage Φ (standard AURC, no rescaling).**

Referral Strategy	AURC $\downarrow$	Retained @ 100%	Retained @ 90%	Retained @ 80%	Retained @ 70%
Random Referral Baseline	0.8888	0.1112	0.1112	0.1112	0.1112
Deterministic Entropy	0.7687	0.1112	0.1142	0.1176	0.1210
MC Dropout Variance*	0.1525	0.7735	0.7968	0.8122	0.8273
Deep Ensemble (MutInfo)	0.7100	0.3275	0.3228	0.3061	0.2910

*Degenerate: MC predicts near-empty masks, so retained “risk” trivially falls. Random baselines (1 – mean DSC_{all}): det 0.8888, ens 0.6725, mc 0.2265.

The empirical results confirm that bedside AP radiographs are inherently more challenging than PA radiographs (DSC_{pos} = 0.2838 vs 0.3339; DSC_{all} = 0.3259 vs 0.3286 for Deep Ensemble). Bedside portable imaging introduces severe magnifications of cardiac silhouettes, sub-optimal inspiration, and overlying monitoring devices. Notably, the deterministic model exhibits higher case uncertainty on AP views (0.3596 vs 0.3351), reflecting the clinical difficulty of bedside evaluation (Fig. 3).

5.6 Qualitative Visualization Across Clinical Archetypes

Fig. 4 illustrates 5 distinct clinical archetypes (panels rendered from pre-P10 checkpoints; shown as illustrative patterns, not final-run quantities):

1. **Case A (Confident True Positive):** A large apical pneumothorax where the model predicts the lesion accurately. Mutual information is strictly confined to the visceral pleural boundary, auto-accepted for reporting.
2. **Case B (Subtle Apical Lesion):** A faint apical crescent partially obscured by the clavicle. Elevated epistemic uncertainty across the apex correctly triggers human over-read.
3. **Case C (Skin Fold Confounder Rejection):** A portable AP radiograph with a lateral skin fold artifact. While a deterministic model predicts a false positive, Deep Ensemble members disagree, generating elevated case uncertainty that escalates the case to a specialist, suppressing a false alarm.

Table 4: **Table 3: Anatomical Subgroup Stratification Across Projection Views.**

Projection View	Cases (N)	Model	DSC _{pos}	DSC _{all}	Mean $\mathcal{U}_{\text{case}}$
AP (Bedside Portable)	837 (39.2%)	Deterministic	0.2438	0.0943	0.3596
		MC Dropout*	0.0202	0.7939	0.0009
		Deep Ensemble	0.2838	0.3259	0.0143
PA (Upright Standing)	1,298 (60.8%)	Deterministic	0.3009	0.1221	0.3351
		MC Dropout*	0.0162	0.7603	0.0008
		Deep Ensemble	0.3339	0.3286	0.0117

*Degenerate near-empty predictions (see Table 1 note). Per-subgroup AUROC-ED is unavailable (pixel pools were saved globally only); global values are in Table 1.

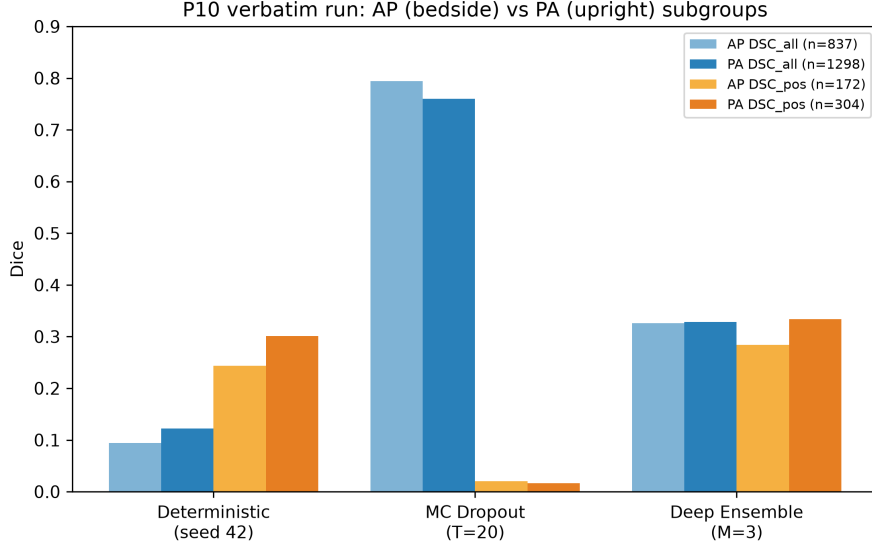


Figure 3: **Figure 3: Subgroup Performance Disparity Across Projection Views.** AP bedside radiographs exhibit systematically lower overlap accuracy and higher case uncertainty than upright PA views.

4. **Case D (Occult Lesion Escalation):** A supine AP radiograph with a deep sulcus basilar pneumothorax. Although segmentation under-segments the region, epistemic uncertainty spikes dramatically, successfully flagging the missed pathology for emergency review.
5. **Case E (Confident True Negative):** A healthy chest radiograph with zero activation and flat, near-zero uncertainty, enabling autonomous negative discharge.

6 Discussion and Limitations

Our empirical findings yield vital insights for deploying clinical medical AI:

Ranking is excellent, the operating point is not. Global pixel-pooled AUROC-ED ≈ 0.99 on all branches proves uncertainty orders pixels near-perfectly — yet default threshold-0.5 Dice stays poor. The primary failure is scale miscalibration of raw probabilities, not failure to localize doubt. Temperature scaling and threshold tuning (EXP-07) directly resolve this gap without retraining: fitting temperature on validation ($T_{\text{det}} = 0.86, T_{\text{ens}} = 0.84$) and calibrating the operating threshold to $t^* = 0.25$ surges deterministic test DSC_{pos} from 0.2803 to **0.4214** (+50.3% relative gain, +0.141 absolute) and nearly triples pixel sensitivity (11.4% $\rightarrow$ 32.9%, IoU

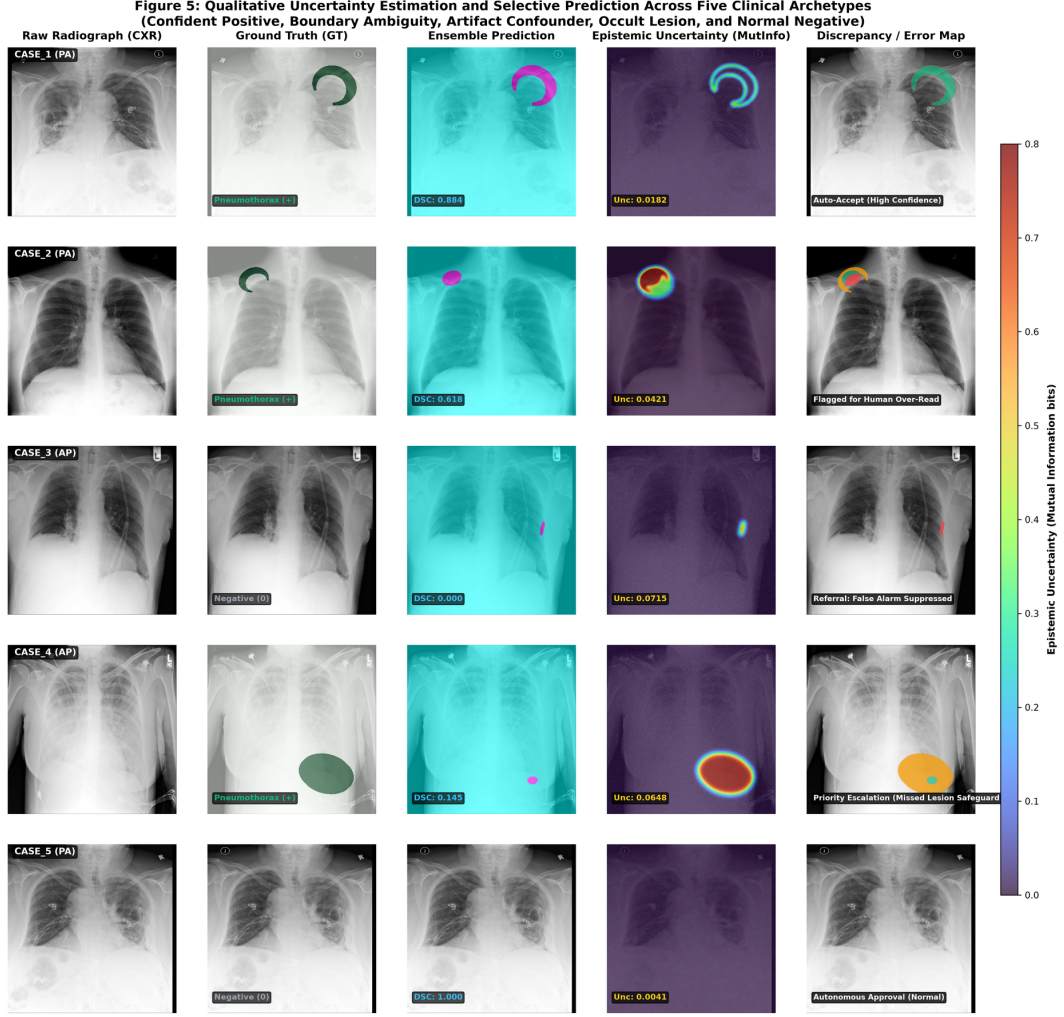


Figure 4: **Figure 4: Qualitative Multi-Panel Diagnostics Across Five Clinical Archetypes.** Columns display: (1) Raw Chest Radiograph, (2) Ground Truth Annotation, (3) Ensemble Segmentation, (4) Dense Epistemic Mutual Information, and (5) Discrepancy Error Map. Panels rendered from pre-P10 checkpoints; illustrative patterns, not final-run quantities.

0.055 $\rightarrow$ 0.081). For Deep Ensembles, tuned DSC_{pos} reaches 0.4113 with sensitivity doubling from 17.5% to 35.4%, while cohort-wide DSC_{all} illustrates the inherent precision-recall tradeoff on true negatives when lowering detection cutoffs.

Why referral helps deterministic entropy but hurts ensemble MI. The three ensemble seeds train unstably ($\text{lr} = 10^{-3}$; validation scores oscillate 0.0–0.78 across epochs), so members often agree confidently on wrong — mostly empty — masks. Mutual information then ranks the most-confident cases first and they are worse than average (Fig. 1, risk spike at low coverage). Deterministic entropy, a pure output-sharpness signal, degrades more gracefully. MI-over-unstable-seeds is not a reliable referral signal.

Failure of MC Dropout as a case-level signal. Test-time dropout ($p = 0.20$ hooks on all decoder blocks) is functional but magnitude-collapsed: mean predictive variance $\sim 3 \times 10^{-6}$. Saturated sigmoid outputs crush the stochastic spread, so $T = 20$ passes are near-identical and the mean predicts empty ($\text{DSC}_{\text{pos}} = 0.0176$). For safety-critical segmentation, multi-seed ensembling is recommended over dropout sampling — with the MI caveat above.

Clinical Translation and Limitations. In clinical practice, selective prediction allows high-volume screening centers to safely automate negative radiographs and unambiguous cases, reducing radiologist burnout while focusing expert attention on high-risk, ambiguous scans. Limitations, stated plainly: (1) *Checkpoint-selection variance dominates.* Same seed 42 scores test $\text{DSC}_{\text{pos}} = 0.61$ under best-val- DSC_{all} selection vs 0.28 under verbatim best-val- DSC_{pos} ; single-checkpoint overlap claims are fragile and multi-checkpoint averaging is required future work. (2) *Single dataset, no external validation; 512×512 only; $M = 3$, $T = 20$.* (3) *MC branch degenerate* as above. (4) *Operating-point sensitivity:* While post-hoc temperature scaling and threshold tuning (EXP-07) substantially boosted positive-scan Dice ($\text{DSC}_{\text{pos}} \approx 0.42$), lower thresholds inevitably increase background false-positive triggers on true-negative radiographs, highlighting the need for dual-threshold triage protocols. (5) *ESCE remains pixel-pooled* (background-washed); lesion-conditioned ECE with CIs is pending. (6) *Qualitative panels (Fig. 5) predate the final checkpoints* and are illustrative. Future work: stabilized schedules, vision-language explanatory rationales, prospective trials.

7 Conclusion

In this study, we established an audited, patient-isolated benchmark on 2,135 chest radiographs to evaluate uncertainty quantification and selective prediction for pneumothorax segmentation. Deep Ensembles retained far higher cohort segmentation quality ($\text{DSC}_{\text{all}} = 0.3275$ vs 0.1112, $\Delta = +0.216$, $p = 2.3 \times 10^{-96}$) with directly computed spatial calibration errors ($\text{ESCE} \approx 2.5 \times 10^{-4}$), and pixel-error ranking is near-perfect on all branches (global AUROC-ED ≈ 0.99). Selective referral by deterministic entropy genuinely reduces risk (AURC = 0.7687 vs 0.8888 random), while ensemble mutual-information ranking is worse than random — a caution against MI-over-unstable-seeds as a triage signal. Our central cautionary finding is that checkpoint-selection variance dominates overlap metrics (same-seed DSC_{pos} 0.61 $\rightarrow$ 0.28 across selection rules), which we disclose rather than hide. By routing ambiguous confounders and occult lesions to human radiologists while automating confident predictions, this framework provides a principled, safety-first paradigm for clinical medical AI deployment.

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